0000-0002-5340-5154

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
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NAME: Ehrlinger, John

eRA COMMONS USER NAME (credential, e.g., agency login): EHRLINJ

POSITION TITLE: Assistant Staff, Lead Data Scientist

EDUCATION/TRAINING (*Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.*)

INSTITUTION AND LOCATION	DEGREE (*if applicable*)	Completion Date MM/YYYY	FIELD OF STUDY
Case Western Reserve University, Cleveland, OH	BS	05/1993	Mechanical Engineering
Case Western Reserve University, Cleveland, OH	MS	05/1994	Mechanical / Aerospace Engineering
Case Western Reserve University, Cleveland, OH	PhD	05/2011	Statistics

A. Personal Statement

I am an applied statistician and data scientist with expertise in computational statistical machine learning, deep learning, and their application to clinical and engineering data. My research career spans both the theoretical development of machine learning methods and their direct application to health outcomes, with a particular focus on time-to-event data, longitudinal and temporal data analysis, and survival methods.

My foundational methodological work on regularization and gradient boosting—specifically the theoretical characterization of ℓ_2Boosting and its connections to the lasso and stagewise regression—established a principled framework for understanding regularized machine learning in high-dimensional settings. I have extended this work to multivariate longitudinal settings through boosted tree ensembles, enabling richer modeling of complex clinical trajectories. In parallel, I have developed open-source software tools (ggRandomForests in R; hazard in SAS/C) that make these methods accessible to the broader statistical and biomedical research community.

I have since returned to Cleveland Clinic's Heart, Vascular & Thoracic Institute as Assistant Staff, Lead Data Scientist in the Department of Cardiothoracic Surgery, where I lead a team of data engineers and data scientists focused on cardiovascular outcomes research. In this role I drive statistical methods research as applied to observational clinical research, establish best practices in software engineering and reproducible research, and implement process improvements to optimize the departmental research pipeline from data collection through publication.

Selected citations relevant to this application:

1. **Ehrlinger J**, Ishwaran H. Characterizing ℓ_2Boosting. *The Annals of Statistics.* **40**(2): 1074–1101. 2012.
2. Pande A, Li L, Rajeswaran J, **Ehrlinger J**, Blackstone EH, Ishwaran H, Lauer MS. Boosted multivariate trees for longitudinal data. *Machine Learning.* **106**(2): 277–305. 2017.
3. Hurst T, Xanthopoulos A, **Ehrlinger J**, et al. Dynamic prediction of LVAD pump thrombosis based on lactate dehydrogenase trends. *ESC Heart Failure.* **6**(5): 1005–1014. 2019.
4. **Ehrlinger J**. ggRandomForests: Exploring Random Forest Survival. *arXiv preprint* arXiv:1612.08974 [stat.CO]. 2016.

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

[t]X[1,l]X[6.5,l,p] 1999–2012 Lead Systems Analyst; Scientific Programmer, Dept. of Thoracic and Cardiovascular Surge

an open-source implementation of multi-phase hazard analysis for time-to-event decomposition originally developed at Cleveland Clinic.

a. **Ehrlinger J**. ggRandomForests: Visually Exploring a Random Forest for Regression. *arXiv preprint* arXiv:1501.07196 [stat.CO]. 2015.
b. **Ehrlinger J**. ggRandomForests: Exploring Random Forest Survival. *arXiv preprint* arXiv:1612.08974 [stat.CO]. 2016.

3. **Temporal and Longitudinal Modeling for Cardiac Outcomes.**

A core application of my methodological work has been the modeling of time-varying clinical data from cardiac patients, including parametric nonlinear temporal decomposition models and mixture approaches to characterize atrial fibrillation recurrence and prevalence.

a. Rajeswaran J, Blackstone EH, **Ehrlinger J**, Thuita L, et al. Probability of atrial fibrillation after ablation: Using a parametric nonlinear temporal decomposition mixed effects model. *Statistical Methods in Medical Research.* **27**(1): 126–141. 2018.
b. Li L, Mao H, Ishwaran H, **Ehrlinger J**, et al. Estimating the prevalence of atrial fibrillation from a three class mixture model for repeated diagnoses. *Biometrical Journal.* **59**(2): 331–343. 2017.
c. Blackstone EH, Chang HL, Rajeswaran J, **Ehrlinger J**, et al. Biatrial maze procedure versus pulmonary vein isolation for atrial fibrillation during mitral valve surgery. *The Journal of Thoracic and Cardiovascular Surgery.* **157**(1): 234–243.e9. 2019.
d. Hurst T, Xanthopoulos A, **Ehrlinger J**, et al. Dynamic prediction of LVAD pump thrombosis based on lactate dehydrogenase trends. *ESC Heart Failure.* **6**(5): 1005–1014. 2019.

4. **Machine Learning Applications in Mechanical Circulatory Support.**

My clinical collaboration work at Cleveland Clinic's Heart and Vascular Institute yielded several high-impact contributions identifying risk factors and predictive biomarkers for LVAD pump thrombosis. The 2014 NEJM paper documenting an abrupt increase in LVAD thrombosis rates had immediate regulatory and clinical implications nationally.

a. Starling RC, Moazami N, Silvestry SC, Ewald G, Rogers JG, Milano CA, Rame JE, Acker MA, Blackstone EH, **Ehrlinger J**, et al. Unexpected Abrupt Increase in Left Ventricular Assist Device Thrombosis. *New England Journal of Medicine.* **370**(1): 33–40. 2014.
b. Smedira NG, Blackstone EH, **Ehrlinger J**, Thuita L, Pierce CD, Moazami N, Starling RC. Current risks of HeartMate II pump thrombosis. *The Journal of Heart and Lung Transplantation.* **34**(12): 1527–1534. 2015.
c. Wojnarski CM, Roselli EE, Idrees JJ, et al. Machine-learning phenotypic classification of bicuspid aortopathy. *The Journal of Thoracic and Cardiovascular Surgery.* **155**(2): 461–469.e4. 2018.

5. **Cardiovascular Surgical Outcomes and Decision Support.**

Spanning my career at Cleveland Clinic, I have contributed analytical expertise to outcomes studies covering LVAD support, transcatheter aortic valve replacement (TAVR/PARTNER trial), and esophageal cancer surgery.

a. Szeto WY, Svensson LG, Rajeswaran J, **Ehrlinger J**, et al. Appropriate Patient Selection or Healthcare Rationing? Lessons from Surgical Aortic Valve Replacement in the PARTNER-I Trial. *The Journal of Thoracic and Cardiovascular Surgery.* **150**(3): 557–568. 2015.
b. Raja S, Rice TW, **Ehrlinger J**, et al. Importance of Residual Cancer after Induction Therapy for Esophageal Adenocarcinoma. *The Journal of Thoracic and Cardiovascular Surgery.* **152**(3): 756–761. 2016.
c. Dalton JE, Glance LG, Mascha EJ, **Ehrlinger J**, Chamoun N, Sessler DI. Impact of Present-on-admission Indicators on Risk-adjusted Hospital Mortality Measurement. *Anesthesiology.* **118**(6): 1298–1306. 2013.
d. Wojnarski CM, Svensson LG, Roselli EE, et al. Aortic Dissection in Patients with Bicuspid Aortic Valve–Associated Aneurysm. *Annals of Thoracic Surgery.* **100**(5): 1666–1674. 2015.

D. Additional Information: Research Support and/or Scholastic Performance

Pending Research Support

NIH – R01 (submitted Feb 2026) 2026– N/A
Improving Waitlist and Transplant Survival in Advanced Heart Failure Populations
Multiple PIs: Eileen Hsich, M.D. and Eugene H. Blackstone, M.D.
Co-Investigator

Completed Research Support

NIH – 1R01HL103552-01A1 8/2011–5/2014 N/A
Ancillary Comparative Effectiveness of Atrial Fibrillation Ablation Surgery
P.I.: Eugene H. Blackstone, M.D.
Co-Investigator
State of Ohio – Wright Center of Innovation 8/2005–12/2010 N/A
Atrial Fibrillation Innovation Center State of Ohio
P.I.: A. Marc Gillinov, M.D.
Scientific programming and computing systems management
NIH – HHSN26820080026C 9/2008–9/2010 N/A
NHLBI Quantitative Clinical Cardiovascular Epidemiology Projects
P.I.: Eugene H. Blackstone, M.D.
Scientific programming and computing systems management
NIH – 5R01HL072771 10/2005–9/2008 N/A
Logical Analysis of Data and Cardiac Surgery Risk
P.I.: Michael S. Lauer, M.D.
Scientific programming and computing systems management